

Managing the Test Process

Testing isn't just a skill — it's a project. Someone has to plan it, staff it, and steer it to a finish.

Test Management · Test Organization · Team Structure · Test Planning

WHY THIS TOPIC EXISTS

Knowing HOW to test isn't enough — someone has to decide what to test, when, by whom, and whether you're done.

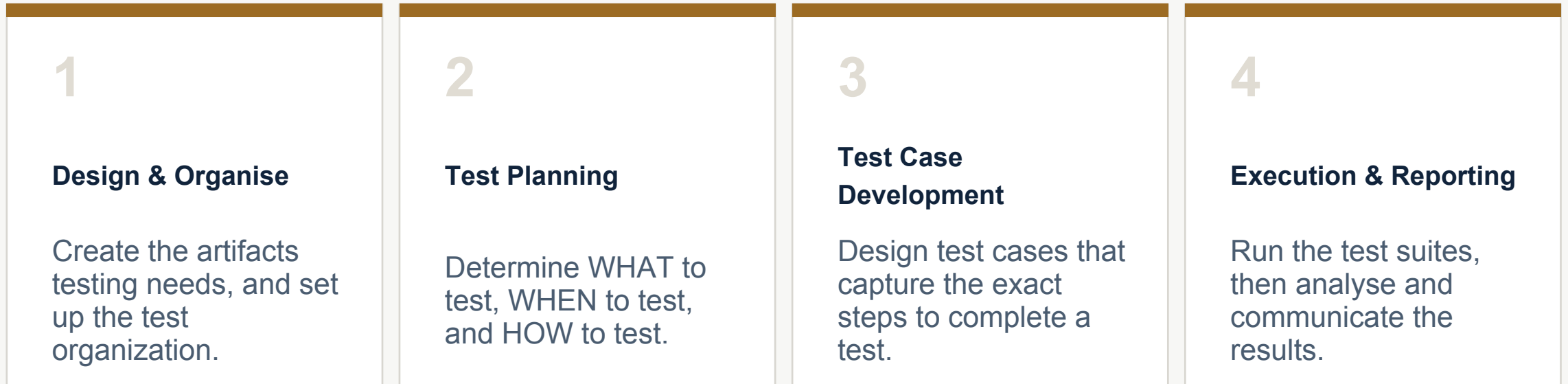
So far we've studied testing techniques — the tools. But on a real project, testing is an activity that must be organised and controlled: roles assigned, resources allocated, risks anticipated, and a plan written down. Test management is the practice of doing exactly that, so a whole team can plan, develop and execute testing across the entire software development process.

The goal: ensure software quality by defining, measuring and tracking quality goals — as a managed project.

What Is Test Management?

Test management defines the practices followed to organise and control the process of testing — and to develop the artifacts testing needs. It makes sure those artifacts are available and easy to access for the whole team.

THE FOUR PHASES OF TEST MANAGEMENT



Challenges in Test Management

Why testing needs active management — these are the pressures every test manager faces.

The four questions

What to test, why, when and how — rarely obvious, always contested.

Not enough time

Delays upstream squeeze the testing phase and pile pressure on the team.

Too few resources

The effort a complete system needs usually exceeds what's available.

Distributed teams

Time-zone and cultural differences make coordination harder.

Unstable requirements

Frequently changing, ambiguous or incomplete requirements are the norm.

Dev & test out of sync

The two teams should work in step — but often don't.

Test Management Recommendations

The textbook's advice for managing testing well — the antidote to the challenges we just saw.

- ✓ Start test management activity EARLY.
- ✓ Reuse test artifacts wherever possible.
- ✓ Introduce testing resources across locations.
- ✓ Coordinate and communicate with the development team.
- ✓ Treat testing as an iterative process.
- ✓ Perform requirement-based testing.
- ✓ Define a repeatable AND flexible testing process.
- ✓ Stay focused on goals and results.

SECTION 3.1.1

Test Organization

Before you test, you decide WHO does it. Five setup tasks turn a vague intention into a real testing team.

Five Tasks of Test Organization

Test organization defines the roles, responsibilities, functions and facilities of the testing process — and the competencies of the people involved. It is done in five tasks.

1	Appoint the Test Manager	Decide who leads testing and is accountable for its success.
2	Define the Scope of Testing	Establish what will — and won't — be tested, before the plan.
3	Appoint the Test Team	Form the team and clearly define each member's role.
4	Verify Development Documentation	Ensure all documents testing needs are available to the team.
5	Validate the Test Estimate	Check the test estimates and the project status process.

The Right Manager for the Size

A test manager builds and leads the team, defines the testing scope per release, deploys resources, applies metrics, and manages the effort. But which skills matter most depends on the project's size:

SMALL

2–3 testers

Good TESTING experience matters most — the manager is hands-on.

MEDIUM

3–6 testers

BOTH testing and management experience are needed.

LARGE

6+ testers

MANAGEMENT skills matter more than testing skills.

Four Ways to Build a Test Team

Who should test? Each staffing approach trades cost against independence.

Approach	Who	Advantages	Disadvantages
Internal	Project team	Low cost, knows the project	Lacks independence & objectivity
External	QA / pro testers	Independent, multi-project experience	Costly, over-reliance
Non-IT	Real users	Independent, real-user view	Lacks IT & project knowledge
Combination	Any / all above	Multiple skills & perspectives	Cost, harder to schedule

Rule of thumb: the more independent the tester, the more objective the result — but usually the higher the cost.

Why You Shouldn't Test Your Own Work

The textbook warns: when a person checks only their OWN work, blind spots creep in. This is the deeper reason testing is often a separate role.

Misunderstandings go uncorrected

If they misread a requirement, there's no one to catch it — they carry on with the wrong facts.

Process errors stay invisible

By their own logic they ARE following the right process, so mistakes in it aren't recognised.

Wrong results get accepted

The same misunderstanding that caused the defect makes them accept the wrong result.

Over-confidence

Believing too much in their own defect-free work, they don't test extensively.

SECTION 3.2

Testing Team Structure

A test team is a cast of specific roles — from the manager who steers, down to the administrator who keeps the lab running.

The Five Roles of a Test Team

1	Test Manager	Plans and executes the project, defines scope, manages schedule and risk, applies metrics, makes the critical decisions.
2	QA Lead	Mentors the team, implements industry best practice, develops QA processes and documents, runs day-to-day QA activity.
3	Test Analyst	Reads and analyses requirement & design documents, executes tests, gathers and manages test data, evaluates outcomes.
4	Test Engineer	Writes and executes test cases, reports defects, and finds the best way to reach 100% test coverage.
5	Test Administrator	Builds, manages and maintains the test environment and assets, and supports the team in using it for execution.

What Makes a Team Effective

Roles alone don't make a good team. The textbook lists four human qualities that turn a group of testers into an effective one.

Strong Cooperation

Working with others toward a shared job solving interpersonal issues and supporting each other.

Commitment

Every member commits to the common objectives and takes responsibility for their own and the team's work.

Effective Communication

Members speak up and express ideas clearly, honestly and logically, so everyone understands each other.

Sharing

Everyone has strong and weak points; a good team shares knowledge and experience to skill each other up.

SECTION 3.3

Test Planning

The heart of managing testing: five steps that turn 'we should test this' into a written, agreed plan.

Test Planning in Five Steps

1	Analyze the software project	Understand the project's attributes you need more than the dev plan contains.
2	Understand the project's risks	Identify what could go wrong, then plan to reduce its impact (risk mitigation).
3	Select a testing technique	Choose the techniques and the test method — dynamic or static to implement them.
4	Plan unit testing & analysis	Verify features against the spec (unit testing) and detect issues without running code (analysis).
5	Build the test plan	Write the document that describes the scope, approach, resources and schedule.

Build a Complete Project Profile

Effective planning is only possible when testers understand the software being tested. Because testing starts early, the project plan may not even be complete yet — so testers gather a fuller picture, including the client's and end-user's expectations.

A PROJECT PROFILE CAPTURES...

- Project objectives

- Development process used

- Customers / users of the software

- Cost & schedule

- Project constraints

- Legal / industry issues

- Technologies implemented

Risk & Risk Mitigation

Risk = a future, UNCERTAIN event with a probability of occurrence and a potential for loss.

EXAMPLE RISKS

- Team lacks the skills the testing needs
- The schedule is too tight to finish on time
- The test manager has poor management skill
- Lack of cooperation hurts productivity
- Wrong budget estimate → cost overruns

RISK MITIGATION

- ✓ Plan training to skill up members
- ✓ Set a test PRIORITY for each activity
- ✓ Plan leadership training for the manager
- ✓ Encourage & inspire each team member
- ✓ Establish scope early; track progress

Choose the Technique, Plan the Units

STEP 3 · SELECT A TESTING TECHNIQUE

Pick techniques by their ability to meet the test objectives, then decide the test METHOD to implement them — dynamic (run the program) or static (examine without running).

Types available include:

Requirements · Regression · Stress · Recovery · Security
· Control · Parallel · Compliance testing

STEP 4 · PLAN UNIT TESTING & ANALYSIS

Unit testing verifies that the program's features work as per the specification — test cases run against the module to check it gives the expected output.

*Analysis is a STATIC approach — required features are detected by examining, not executing, the code.
Techniques: review, inspection, audit.*

The Test Plan Document

A test plan is a document describing the scope, approach, resources and schedule of the intended testing activities — identifying the test items, the features to test, the tasks, who does each, and any risks needing contingency planning.

WHY IT MATTERS

A single source of truth

Everyone testers, developers, managers, client reads the same agreed scope and approach.

It answers 'are we done?'

Pass/fail criteria and deliverables make completion objective, not a matter of opinion.

It makes testing accountable

Naming who does each task, on what schedule, means no gap is left unowned.

Structure of a Test Plan

Every test document has a unique ID. A standard test plan is organised into these sections:

Introduction

Overview and the goal / objectives of the plan.

Test Items

The releases / versions of the software to be tested.

Features Not to be Tested

What's out of scope and the reason why.

Item Pass/Fail Criteria

How to decide whether a test case passes or fails.

Test Environment

Hardware, software, network, and any tools needed.

References

All documents the plan relies on (e.g. project plan, BRS).

Features to be Tested

The features that ARE in scope.

Approach

Testing level, testing types, and methodology (black/white box).

Test Deliverables

Scripts, cases, defect logs, reports.

Estimate

Testing cost, resources, time and effort.